Wan Luan Lee

Madison, WI, 53713 | wlee329@wisc.edu | 312-709-0520 | https://wanluanlee.github.io

EDUCATION

University of Wisconsin – Madison | Madison, WI — ***July 2026 (expected)***

Ph.D. Electrical and Computer Engineering

- GPU-accelerated regular and hypergraph partitioning
- GPU-accelerated incremental graph partitioning
- Large Language Model-powered code generation assistant for graph partitioning workloads

Brigham Young University | Provo, UT — ***June 2017 – Aug 2020***

Bachelor of Science in Chemical Engineering

- GPA: 3.7/4.0
- Accurately predicted the behavior and structural properties of molybdenum solvated in fluoride salt using molecular simulations

EXPERIENCE

UW – Madison Department of Electrical and Computer Engineering | Madison, WI — ***Jan 2022 – Present***

Research Assistant

- Developed a GPU-based graph partitioning algorithm, achieving a **9x** speedup over state-of-the-art CPU partitioners
- Implemented a GPU-accelerated hypergraph partitioning algorithm, delivering a **4x** performance boost compared to leading CPU tools
- Designed a GPU-based incremental hypergraph partitioner, enabling real-time partition updates with minimal computational overhead

Cadence Design Systems | San Jose, CA — ***May 2025 – Aug 2025 (expected)***

Summer Intern

- Developed GPU-accelerated solutions to enhance the performance of emulation workloads, targeting critical bottlenecks in existing hardware acceleration pipelines
- Benchmarked GPU-based accelerators against CPU baselines and proposed optimization strategies based on performance profiling

UW – Madison Department of Electrical and Computer Engineering | Madison, WI — ***Sep 2024 – May 2025***

Teaching Assistant

- Assisted in teaching a 100-student course focused on parallel programming with OpenMP and CUDA
- Conducted weekly office hours and discussion sessions to support students on performance tuning, debugging, and understanding parallel computing concepts
- Provided one-on-one feedback and guidance to students on their final HPC projects, evaluating proposed methods for feasibility, performance potential, and alignment with course principles.

Synopsys | Mountain View, CA — ***May 2022 – July 2022***

Summer Intern

- Conducted performance analysis on the Synopsys timing tool to identify bottlenecks
- Implemented GPU-accelerated solutions for heavy computations in timing analysis, achieving significant performance improvements

Secumax Technology Solution| Taipei, Taiwan — ***Sep 2020 – May 2021***

Software Developer

- Designed and implemented a SQL database to store and handle application data efficiently and securely
- Utilized Qt to create a cross-platform application that allows users to control remote monitors and perform features such as zoom and playback

- Worked with the sales team to design and implement a customer-driven user interface, enhancing the user experience

WeCare Insurance Company | Provo, UT ***Apr 2020 – Aug 2020***
Summer Intern
- Led a team of three, mentoring members and coordinating tasks to ensure timely project delivery
- Developed web automation scripts using Selenium and JavaScript to streamline the generation of insurance quotes, improving efficiency and accuracy
- Optimized SQL schema to improve query performance and streamline data access for downstream applications

Brigham Young University Department of Chemical Engineering | Provo, UT ***Jan 2020 – Apr 2020***
Teaching Assistant
- Conducted weekly office hours to support 45 students in understanding core Environmental Safety concepts
- Led review sessions to reinforce key course material and improve student performance

Brigham Young University Department of Chemical Engineering | Provo, UT ***Sep 2017 – Jan 2020***
Research Assistant
- Developed Python and C++ scripts to generate complex formatted files from data
- Automated 100+ LAMMPS molecular simulations on a supercomputer using Bash scripting, streamlining workflow and reducing manual effort
- Evaluated molten salt structure for nuclear reactors through molecular dynamics simulation

PUBLICATIONS

- **Wan Luan Lee**, Dian-Lun Lin, Shui Jiang, Cheng-Hsiang Chiu, Yibo Lin, Bei Yu, Tsung-Yi Ho, and Tsung-Wei Huang. "*G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner using Task Graph Parallelism.*" Accepted to ACM Transactions on Design Automation of Electronic Systems (TODAES) 2025.

- **Wan Luan Lee**, Shui Jiang, Dian-Lun Lin, Che Chang, Boyang Zhang, Ulf Schlichtmann, Tsung-Yi Ho, and Tsung-Wei Huang. "*iG-kway: Incremental k-way Graph Partitioning on GPU*." Accepted to Design Automation Conference (DAC) 2025.

- Jie Tong, **Wan Luan Lee**, Umit Yusuf Ogras, and Tsung-Wei Huang. *"Scalable Code Generation for RTL Simulation of Deep Learning Accelerators with MLIR."* Accepted to International European Conference on Parallel and Distributed Computing (Euro-Par) 2025.

- Yi-Hua Chung, Shui Jiang, **Wan Luan Lee**, Yanqing Zhang, Haoxing Ren, Tsung-Yi Ho and Tsung-Wei Huang. *"SimPart: A Simple Yet Effective Replication-aided Partitioning Algorithm for Logic Simulation on GPU."* Accepted to International European Conference on Parallel and Distributed Computing (Euro-Par) 2025.

- **Wan Luan Lee**, Dian-Lun Lin, Cheng-Hsiang Chiu, Ulf Schlichtmann, and Tsung-Wei Huang. "*HyperG: Multilevel GPU-Accelerated k-way Hypergraph Partitioner*." Asia and South Pacific Design Automation Conference (ASP-DAC) 2025.

- Che Chang, Boyang Zhang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yi-Hua Chung, **Wan Luan Lee**, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang. *"PathGen: An Efficient Parallel Critical Path Generation Algorithm."* Asia and South Pacific Design Automation Conference (ASP-DAC) 2025.

- Boyang Zhang, Che Chang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, **Wan Luan Lee**, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang. *"iTAP: An Incremental Task Graph Partitioner for Task-parallel Static Timing Analysis."* Asia and South Pacific Design Automation Conference (ASP-DAC) 2025.

- Boyang Zhang, Dian-Lun Lin, Che Chang, Cheng-Hsiang Chiu, Bojue Wang, **Wan Luan Lee**, Chih-Chun Chang, Donghao Fang, and Tsung-Wei Huang. "*G-PASTA: GPU Accelerated Partitioning Algorithm for Static Timing Analysis."* Design Automation Conference (DAC) 2024.

- **Wan Luan Lee**, Dian-Lun Lin, Tsung-Wei Huang, Shui Jiang, Tsung-Yi Ho, Yibo Lin, and Bei Yu. *"G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner*." Design Automation Conference (DAC) 2024.

- Austin David Clark, **Wan Luan Lee**, Andrew Russell Solano, Tyler Bruce Williams, Gabriel Scott Meyer, Granite J Tait, Ben C Battraw, and Stella D Nickerson. "*Complexation of Mo in FLiNaK molten salt: insight from ab initio molecular dynamics*." The Journal of Physical Chemistry B 2020.

AWARDS

- 2023 Design Automation Conference Young Fellowship
- 2024 Design Automation Conference Young Fellowship
- Brigham Young University Academic Scholarship
- Brigham Young University Chemical Engineering Department Scholarship